A Survey of Perceptual Hashing for Multimedia

YUANDING ZHOU, School of Optical-Electrical and Computer Engineering, University of Shanghai for Science and Technology, Shanghai, China
XINRAN LI, Business School, University of Shanghai for Science and Technology, Shanghai, China
CHENG XIONG, HENG YAO, and CHUAN QIN, School of Optical-Electrical and Computer Engineering, University of Shanghai for Science and Technology, Shanghai, China

Perceptual hashing is a cutting-edge technique in the field of digital multimedia security, which maps the perceptual content of multimedia information to a fixed-length hash sequence to achieve content authentication. This survey provides a systematic overview of the definition, basic steps, main characters, and application scenarios of perceptual hashing. According to the different authentication objects, representative schemes of perceptual image hashing, perceptual video hashing, and perceptual hashing for neural network models are introduced, respectively. Both perceptual image hashing and perceptual video hashing can be divided into classical methods-based schemes and learning-based schemes, where learning-based schemes can be subdivided into supervised and unsupervised. Classical methods-based image hashing schemes can be divided into four categories: local feature-based, transformation-based, statistical feature-based, and dimensionality reduction-based. Classical methods-based video hashing schemes are mainly categorized into spatial feature-based schemes and spatial-temporal feature-based schemes. Additionally, we introduce the dataset composition and hash distance metric strategies for perceptual hashing and analyze the performance of some representative schemes. Finally, we summarize the existing schemes and offer prospects for future research directions and development trends.

CCS Concepts: • **Security and privacy** → **Security services**; **Authentication**;

Additional Key Words and Phrases: Perceptual hashing, Robustness, Discrimination, Security, Multimedia

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Authors' Contact Information: Yuanding Zhou, School of Optical-Electrical and Computer Engineering, University of Shanghai for Science and Technology, Shanghai, China; e-mail: 221240081@st.usst.edu.cn; Xinran Li, Business School, University of Shanghai for Science and Technology, Shanghai, China; e-mail: xinranli@usst.edu.cn; Cheng Xiong, School of Optical-Electrical and Computer Engineering, University of Shanghai for Science and Technology, Shanghai, China; e-mail: 221240066@st.usst.edu.cn; Heng Yao, School of Optical-Electrical and Computer Engineering, University of Shanghai for Science and Technology, Shanghai, China; e-mail: hyao@usst.edu.cn; Chuan Qin (corresponding author), School of Optical-Electrical and Computer Engineering, University of Shanghai for Science and Technology, Shanghai, China; e-mail: qin@usst.edu.cn.
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1 Introduction

With the development of Internet and multimedia technology, copyright protection is facing unprecedented challenges [1, 2]. Nowadays, multimedia information can be disseminated rapidly and widely, but with the advancement of multimedia editing tools, piracy and copyright infringement can be seen everywhere, yet it is difficult to track. Effective content authentication schemes are urgently needed to ensure the authenticity and security of multimedia information [3, 4].

Perceptual hashing (also called robust fingerprinting or robust hashing) is a semi-active multimedia content authentication technique, which extracts perceptual contents of multimedia information and maps them to a fixed-length hash sequence to achieve content authentication. According to the different objects to be authenticated, it can be generally divided into perceptual image hashing, perceptual video hashing, and perceptual hashing for neural network models, and so forth.

In real-world scenarios, multimedia data often undergo digital processing like compression, enhancement, and format conversion, which results in changes in storage size and data representation compared to the original data, but the perceptual content of these processed multimedia data remains intact. Traditional cryptographic hashing functions (such as SHA-1 and MD5) [5] are highly vulnerable to multimedia content-preserving manipulations such as noise addition, scaling, rotation, compression, and filtering, even when these manipulations have minimal impact on the perceptual content [6]. To overcome this limitation, the concept of perceptual hashing was initially proposed in the early 21st century. Perceptual hashing effectively compresses multimedia data from the perspective of human visuals to generate robust hash sequences. Perceptual hashing has become a key technique in the field of computer vision, showing great application potential.

1.1 Fundamental of Perceptual Hashing

This section offers a comprehensive exploration of perceptual hashing techniques. The basic steps of hash generation and the main characters of hash sequences are described in detail [7].

1.1.1 Basic Steps. The typical structure of perceptual hashing schemes comprises three stages: pre-processing, feature extraction, and hash generation. These steps are visually depicted in Figure 1.

(1) *Pre-Processing*. To augment salient perceptual features of multimedia data, pre-processing is typically applied to the original data prior to subsequent procedures. These pre-processing steps encompass actions such as resizing and noise smoothing, which serve to enhance the quality of multimedia data for further analysis. Common techniques utilized include bilinear interpolation and Gaussian low-pass filtering. Additionally, the color space of the multimedia data can be transformed to facilitate feature extraction. By performing this series of pre-processing operations on the original multimedia data, a secondary representation of multimedia data can be obtained. The main purpose of this step is to facilitate the extraction of perceptual features while mitigating distortions introduced by specific content-preserving manipulations.

(2) *Feature Extraction*. The core step of the perceptual hashing scheme is feature extraction, which endeavors to extract stable features that describes the perceptual content of multimedia data. Various features including texture, color, structure, and other features are extracted from multimedia data to represent the perceptual content. Key of feature extraction is to transform abstract perceptual information into digital features while maintaining its robustness and discrimination capability. This process establishes the foundational requisites for subsequent verification via the corresponding hash sequence during the authentication phase.

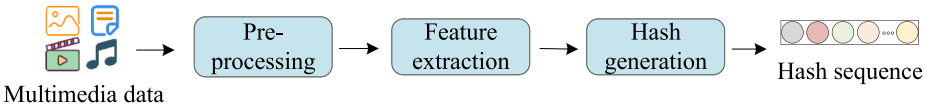


Fig. 1. The main steps of perceptual hashing scheme.

(3) *Hash Generation*. To enhance computational efficiency during hash generation and facilitate the storage of hash sequences, it is imperative to convert perceptual features into a standardized fixed format. Subsequently, pseudo-random keys are employed to perturb, shuffle, and perform additional operations on the intermediate hash sequence, yielding the final hash sequence. This process further enhances the security and discrimination of the final hash sequence.

1.1.2 Main Characters. Let the perceptual identical multimedia data of the original data $\mathbf{I}$ be $\widehat{\mathbf{I}}$ (obtained by enhancement, filtering, or other content-preserving manipulations on $\mathbf{I}$), and let $\widetilde{\mathbf{I}}$ be perceptual distinct multimedia data unrelated to $\mathbf{I}$. The five characters of an ideal perceptual hashing scheme $\mathcal{H}(\cdot)$ are as follows:

(1) *One-Way*. A perceptual hashing scheme facilitates the unidirectional mapping of multimedia data content to a hash sequence. However, it is important to note that the reverse process of deriving the original perceptual content from the generated hash sequence is not feasible.

(2) *Perceptual Robustness*. Perceptual robustness is the characteristic that two multimedia data sharing similar perceptual content should yield identical hash sequences, thereby necessitating resilience against common content-preserving manipulations. More precisely, the hash sequences of the original multimedia data $\mathbf{I}$ and its corresponding perceptual identical data $\widehat{\mathbf{I}}$ should demonstrate a significant degree of similarity, that is

$$P\{\mathcal{H}(\mathbf{I}) \approx \mathcal{H}(\widehat{\mathbf{I}})\} > 1 - \delta_1. \tag{1}$$

(3) *Discrimination Capability*. Discrimination capability requires that perceptual hashing scheme should be able to effectively distinguish the original multimedia data $\mathbf{I}$ and a perceptual distinct data $\widetilde{\mathbf{I}}$. For two perceptual distinct multimedia data, their hash sequences should be significantly different and the probability of collision between the two hash sequences is required to be almost zero, that is

$$P\{\mathcal{H}(\mathbf{I}) = \mathcal{H}(\widetilde{\mathbf{I}})\} \leq \delta_2. \tag{2}$$

(4) *Key-Dependent Security*. Key-dependent security requires that the generation of hash sequences is controlled by one secret key. With this property, different input keys applied to the same multimedia data will yield different corresponding hash sequences. The probability of generating the same hash sequences from different keys, such as $\mathcal{K}_1$ and $\mathcal{K}_2$, is effectively zero, that is

$$P\{\mathcal{H}_{\mathcal{K}_1}(\mathbf{I}) = \mathcal{H}_{\mathcal{K}_2}(\mathbf{I})\} \leq \delta_3. \tag{3}$$

The $P(\cdot)$ employed in Equations (1)–(3) denotes the probability function, quantifying the likelihood of a specific event occurring. Parameters δ_1, δ_2, and δ_3 represent small positive values.

(5) *Efficiency*. When designing a perceptual hashing scheme, careful consideration must be given to both complexity and compactness. With respect to complexity, the efficiency of

a perceptual hashing scheme is primarily measured by the computational time required to generate a hash sequence for each multimedia data, which should be minimized to the greatest extent possible. Compactness requires that a wide range of features be represented using a hash sequence that is as concise as possible. Striking a balance between complexity and compactness is crucial in constructing an effective perceptual hashing scheme.

1.1.3 Distance Measurement. Due to the diversity in the design of perceptual hashing schemes, the same multimedia data may yield hash sequences in different formats, such as binary, decimal integers, or floating-point numbers. Therefore, it is crucial to choose an appropriate distance measurement method to quantify the relationship between hash sequences. Let two hash sequences of length ℓ be $\mathbf{H}_1 = \{h_1(1), h_1(2), ..., h_1(\ell)\}$ and $\mathbf{H}_2 = \{h_2(1), h_2(2), ..., h_2(\ell)\}$. Details of calculating hash distances between $\mathbf{H}_1$ and $\mathbf{H}_2$ using methods such as Euclidean distance, correlation coefficient, and Hamming distance are as follows:

(1) *Euclidean Distance*. Euclidean distance is a mathematical concept that represents the straight-line distance between two points in a coordinate system. When $\mathbf{H}_1$ and $\mathbf{H}_2$ are in decimal or binary form, Euclidean distance can be used to measure the hash distance, which is given by

$$D_1(\mathbf{H}_1, \mathbf{H}_2) = \sqrt{\sum_{i=1}^{\ell} |h_1(i) - h_2(i)|^2}, \tag{4}$$

where $h_1(i)$ and $h_2(i)$ represent elements of $\mathbf{H}_1$ and $\mathbf{H}_2$, respectively.

(2) *Correlation Coefficient*. Correlation coefficient is a statistical measure that quantifies the linear relationship between variables. Correlation coefficient ranges from $[-1, 1]$, where "-1" indicates a negative correlation, "0" indicates no correlation, and "1" indicates a positive correlation. The hash distance between $\mathbf{H}_1$ and $\mathbf{H}_2$ can be measured by

$$D_2(\mathbf{H}_1, \mathbf{H}_2) = \frac{\sum_{i=1}^{\ell} \left[h_1(i) - \bar{h}_1\right] \left[h_2(i) - \bar{h}_2\right]}{\sqrt{\sum_{i=1}^{\ell} \left[h_1(i) - \bar{h}_1\right]^2} \sqrt{\sum_{i=1}^{\ell} \left[h_2(i) - \bar{h}_2\right]^2}}, \tag{5}$$

where $\bar{h}_1$ and $\bar{h}_2$ are the means of elements in $\mathbf{H}_1$ and $\mathbf{H}_2$, respectively.

(3) *Hamming Distance*. Hamming distance is a measure in information theory that quantifies the number of different characters at corresponding positions between two strings of equal length. It is commonly used to measure the distance between binary hash sequences, and it can be mathematically represented as follows:

$$D_3(\mathbf{H}_1, \mathbf{H}_2) = \sum_{i=1}^{\ell} |h_1(i) \oplus h_2(i)| . \tag{6}$$

(4) *Cosine Similarity*. Cosine similarity evaluates the similarity between two vectors by calculating the cosine value of the angle between them, which has a range of $[0, 1]$. A higher value indicates greater similarity. Cosine similarity is often used for similarity measurement due to its characteristics of being less affected by amplitude and providing high evaluation accuracy, which is given by

$$D_4(\mathbf{H}_1, \mathbf{H}_2) = \sqrt{\frac{(\mathbf{H}_1\mathbf{H}_2)^2}{\left(\mathbf{H}_1{}^{\mathrm{T}}\mathbf{H}_1\right)\left(\mathbf{H}_2{}^{\mathrm{T}}\mathbf{H}_2\right)}}. \tag{7}$$

1.2 Application Scenario

Application scenarios of perceptual hashing are mainly in the field of multimedia security, including copy detection, content authentication, and quality assessment.

1.2.1 Copy Detection. One of the applications of perceptual hashing is the effective copy detection of multimedia data uploaded to the platform to ensure that no duplicate data are stored in the database. By generating a corresponding hash sequence for each newly uploaded multimedia data and comparing it with existing hash sequences in the database, the system can automatically identify duplicate data. Once a duplicate item is detected, corresponding multimedia data are identified and removed from storage. This de-duplication mechanism not only greatly saves valuable storage resources but also further accelerates data retrieval and access by reducing redundant content, and improves the overall retrieval efficiency. It is an effective way to optimize resource utilization and enhance user experience, which enables users to discover and access multimedia data in a clearer and more organized content library.

1.2.2 Content Authentication. With the continuous development of multimedia technology, the need for authentication of multimedia data becomes increasingly prominent. Perceptual hashing emerges as a formidable tool in this endeavor, owing to its robustness and discrimination capabilities. Through the comparison of generated hash sequences, perceptual hashing facilitates the identification and authentication of multimedia data. Notably, the veracity of multimedia data can be determined by assessing the distance between two sequences against a pre-defined threshold [8]. With its multifaceted utility, perceptual hashing finds extensive application in authentication domains, encompassing digital rights protection, copy detection, and tampering detection tasks.

1.2.3 Quality Assessment. Multimedia data often entail various forms of compression during transmission, which leads to a decline in the quality of data [9]. However, users have increasingly higher demands for the quality of multimedia data. Therefore, the assessment of multimedia data quality, especially image quality, is particularly important. Image quality describes the fidelity of image signals, including capture, processing, storage, compression, transmission, and display. The evaluation of image quality can generally be categorized into subjective and objective. Subjective methods rely on the perceptual assessment of image attributes by human observers, while objective methods leverage computational models adept at predicting perceptual image quality [10]. Perceptual hashing generates a set of hash sequences based on image features, which provides a basis for image quality assessment.

1.3 Concept Differentiation

1.3.1 Semantic Retrieval Hashing vs. Perceptual Hashing. Semantic retrieval hashing is a valuable research direction, and cross-modal hashing has attracted considerable attention in recent years [11], including supervised [12] and unsupervised schemes [13]. The focus of semantic retrieval hashing and perceptual hashing is different. Semantic retrieval hashing is to extract semantic information from the query data and compare it with the existing multimedia data in the database to find out results with similar salient regions or the same semantic information [14]. For perceptual hashing, the generated hash sequence of the original multimedia data and perceptual identical data should be as similar as possible, and hash sequence of the original data and perceptual distinct data should be different or have a large hash distance between them. In other words, semantic retrieval hashing utilizes the abstract semantic information of query data to retrieve data with similar semantic content from the dataset. Perceptual hashing aims to identify and authenticate multimedia data in the dataset that have similar perceptual content to a given original data. Figure 2(a) and (b) illustrate similar images in semantic retrieval hashing and perceptual image hashing,



Fig. 2. Similar images in three tasks: semantic retrieval hashing, perceptual image hashing, and near-duplicate detection. Part (a) illustrates similar images in semantic retrieval hashing task, part (b) illustrates identical images in perceptual image hashing task, and part (c) illustrates examples of different near-duplicate images.

respectively. It is important to note that images considered similar in the semantic retrieval hashing task are often perceived as distinct images in perceptual image hashing task.

On the other hand, semantic retrieval hashing focuses on the accuracy and efficiency, which requires matching as many multimedia data as possible in the retrieved dataset that have the same semantic information as the query data. Perceptual hashing task emphasizes the robustness and discrimination capability. Perceptual hashing schemes need to effectively resist the influence of content-preserving manipulations on hash sequences while being able to differentiate between perceptual distinct multimedia data [15].

1.3.2 Near-Duplicate Detection vs. Perceptual Hashing. Near-duplicate detection aims to identify "junk data" (i.e., duplicate data) within a multimedia dataset to reduce data redundancy [16–18]. In near-duplicate detection tasks, near-duplicate multimedia data can be classified into two categories: (1) Identical near-duplicate multimedia data. These data originate from the same digital source and are perceptual identical, but undergo certain transformations or reasonable edits such as noise addition, rotation, minor distortion in brightness, color balance changes, and compression, among others; (2) Different near-duplicate multimedia data. These images come from different digital sources and are captured under different shooting perspectives, lighting conditions, or time periods,

Table 1. The Fields, Sub-Fields, and Schemes of Perceptual Hashing Covered in This Survey

Field	Sub-field		Scheme
Perceptual image hashing	Classical	Local feature	[19–24]
		Transform	[25–29]
		Statistical characteristic	[30–33]
		Dimensionality reduction	[34–40]
	Learning	Supervised	[41–47]
		Unsupervised	[48–51]
Semantic video hashing	Classical	Spatial	[52–59]
		Spatial and temporal	[60–62]
	Learning	Supervised	[63–66]
		Unsupervised	[67, 68]
Perceptual hashing for neural network model			[69–72]

depicting the same scene or object. While the thematic content of these multimedia data is similar, their backgrounds might be different. Figure 2(c) illustrates examples of different near-duplicate images. Near-duplicate detection is similar to semantic retrieval hashing, accuracy serves as the primary metric for near-duplicate detection schemes without emphasizing its robustness against content-preserving manipulations. Compared with perceptual hashing, the definition of identical near-duplicate multimedia data is broader, including perceptual identical multimedia data.

According to the different objects to be authenticated, perceptual hashing can be generally divided into perceptual image hashing, perceptual video hashing, and perceptual hashing for neural network model. As shown in Table 1, these fields can be further divided into several sub-fields. Both perceptual image hashing and perceptual video hashing can be divided into classical methods-based schemes and learning-based schemes, where learning-based schemes can be subdivided into supervised and unsupervised. Classical methods-based image hashing schemes can be divided into four categories: local feature-based, transformation-based, statistical feature-based, and dimensionality reduction-based. Classical methods-based video hashing schemes are mainly categorized into spatial feature-based schemes and spatial-temporal feature-based schemes. The remaining sections of this survey are arranged as follows. Section 2 introduces the classification and representative schemes of perceptual image hashing. Section 3 introduces the representative schemes of perceptual video hashing. Section 4 introduces the perceptual hashing for the neural network model. Section 5 introduces the main performance metrics for evaluating perceptual hashing schemes. Section 6 is the summary of this survey and the prospect of the future.

2 Perceptual Image Hashing

Perceptual image hashing scheme maps perceptual features of an image into a fixed-length hash sequence, which remains robust against image content-preserving manipulations such as rotation, scaling, cropping, and so forth. According to different ways of extracting image features, perceptual image hashing schemes can be divided into classical methods-based and learning-based schemes. Classical methods-based schemes can be further divided into four categories: local feature-based, transformation-based, statistical feature-based, and dimensionality reduction-based. Learning-based schemes can be further divided into supervised and unsupervised perceptual image hashing schemes. Next, we describe the representative perceptual image hashing schemes.

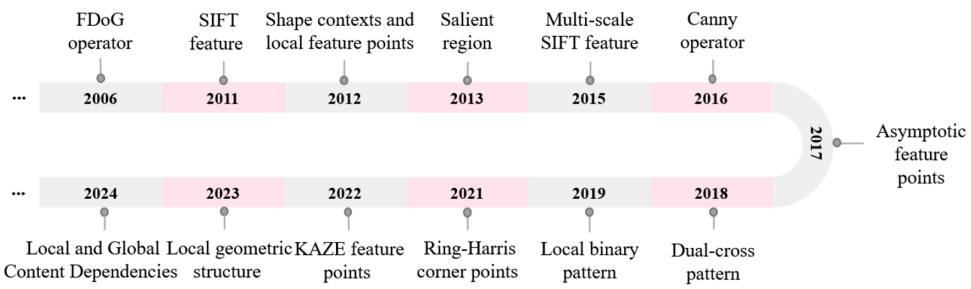


Fig. 3. Development of perceptual image hashing schemes based on local features.

2.1 Classical Methods-Based Perceptual Image Hashing Schemes

2.1.1 Local Feature-Based Schemes. Extensive research has been conducted on local features in addressing robust matching and object detection problems in computer vision [19, 73]. Local feature points, corners, edges, and salient regions have been widely utilized in various computer vision applications such as object detection, recognition, and retrieval [74]. These local feature-based schemes exhibit stable invariance even under geometric distortions [20]. In Figure 3, we present the temporal development of perceptual image hashing schemes based on local features.

Lv et al. [21] proposed a perceptual image hashing scheme by combining **Scale-Invariant Feature Transform (SIFT)** features with the Harris criterion. The introduction of the Harris criterion aimed to enhance the robustness of SIFT keypoint extraction against digital image attacks. The geometric distribution of local feature points and their descriptors were embedded into shape contexts to generate the final hash sequence. Wang et al. [22] utilized the Watson visual model to extract visually sensitive features from images and also combined them with SIFT-based feature extraction to generate intermediate hash sequences. Subsequently, they were dimensionally reduced and scrambled using Gaussian random matrices and encryption randomization operations, respectively, to obtain the final hash sequence. In [20], the SIFT feature vectors of the target image were selectively filtered and compressed. The image summary was then obtained by quantizing the distribution centroid of feature vectors.

The Canny operator [75] can group together pixels in an image that have significant changes in neighborhood grayscale values to form edges of an image. The Canny operator technique has been applied in the works of Qin et al. [23] and Tang et al. [24]. Qin et al. [23] combined the Canny operator with the **Discrete Cosine Transform (DCT)** transform. The pre-processed binary image undergoes edge detection using the Canny operator. The resulting binary edge image was divided into equally sized non-overlapping blocks, and the six blocks with the richest edge features were marked. Each of these six blocks performed DCT processing, and the extracted DCT low-frequency coefficients and block positions were used together as the structural features of the image. Finally, the **Principal Component Analysis (PCA)** dimensionality reduction technique was applied to process structural features and obtain the final hash sequence.

2.1.2 Transform-Based Schemes. Currently, various transform techniques used to extract robust features have been successfully applied in the generation process of perceptual image hashing. These transform techniques generally include **Discrete Fourier Transform (DFT)**, quaternion transform, DCT, Radon transform, **Discrete Wavelet Transform (DWT)**, and log-polar transform, among others. Robust features extracted from the transformed domain exhibit strong resilience against certain image tampering operations [25]. The development of transform-based hashing schemes is illustrated in Figure 4.

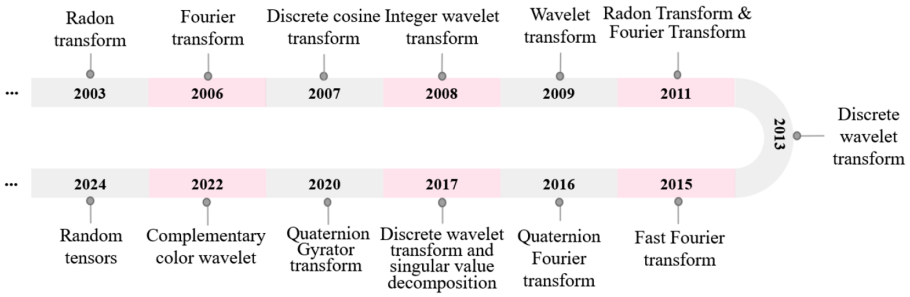


Fig. 4. Development of perceptual image hashing schemes based on transform.

DCT is a decorrelation transform that concentrates energy in low-frequency coefficients, reducing spatial domain redundancy in images while enhancing the robustness of hashing schemes against distortions [76]. In [26], the quadratic image was first divided into non-overlapping image blocks, and then the first row/column of DCT coefficients in each block was selected to construct a feature matrix. In [27], the normalized image was randomly partitioned into overlapping blocks under key control, and the low-frequency DCT coefficients were extracted from each block. The invariant distances between vectors in the quantized feature matrix are calculated, resulting in a compact hash sequence.

Tang et al. [28] divided the quadratic image into non-overlapping blocks and extracted perceptual features using entropy and DWT. For grayscale images, texture features of image blocks were directly represented by their entropy. In the case of color images, the entropies of the R, G, and B components of image blocks were calculated separately, and these entropy values were multiplied and combined. After extracting the entropy, a feature matrix was obtained. To minimize the length of a final hash sequence, the authors performed DWT on the feature matrix and concatenated the coefficients from the LL subband to obtain the final hash sequence. Qin et al. [29] employed a non-linear filtering technique based on total variation [77] to remove noise while preserving significant image edges. The authors generated quadratic images through rotation projection, where lines located at the center of the image were rotated at fixed angles, and the rotated vector's pixels were sequentially projected to form quadratic images. DFT was then applied to the quadratic images, followed by non-uniform sampling from the magnitude matrix of Fourier coefficients to extract robust salient features.

2.1.3 Statistical Characteristics-Based Schemes. Based on the assumption that relative relationships between adjacent pixels remain roughly the same after image distortion, researchers employed statistical features of images to generate hash sequences. Techniques such as image histograms, statistical measures, invariants, and moment features generally exhibit strong robustness against noise, blur, and compression attacks. The development of hashing schemes based on statistical features is depicted in Figure 5.

Due to the fact that image rotation does not alter the pixel clusters within circular ring partitions, many hashing schemes adopt the approach of extracting statistical features from rings to obtain image characteristics. A series of concentric rings are formed by dividing the image based on the distances from pixels to the center of the image. Firstly, the area of an inscribed circle (maximum radius circle) is calculated, and the average area of the inscribed circles is taken as the area for each circular ring. Tang et al. [30] divided the quadratic image into a series of circular rings and used the entropy of each ring to generate a hash sequence. Entropy is a fundamental concept in information theory, and higher entropy values indicate greater randomness of pixels within the

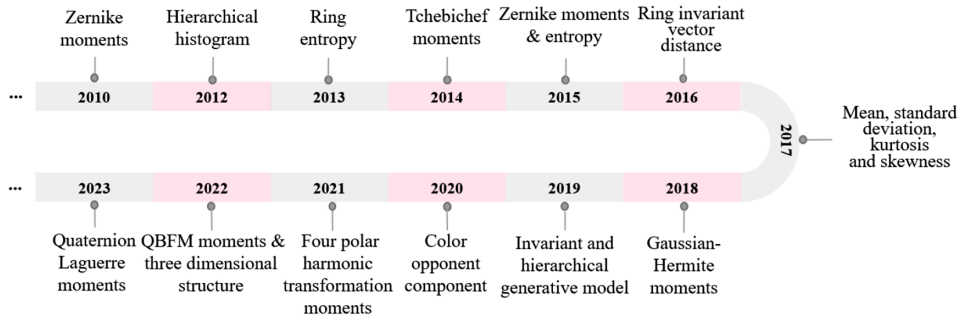


Fig. 5. Development of perceptual image hashing schemes based on statistical characteristics.

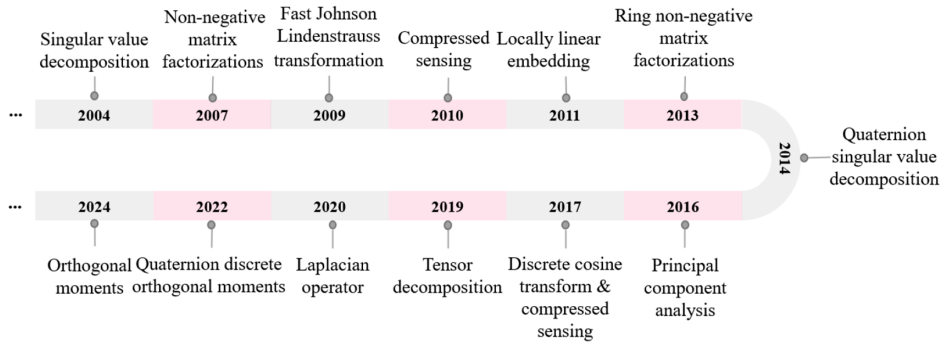


Fig. 6. Development of perceptual image hashing schemes based on dimensionality reduction.

circular ring regions. Subsequently, they proposed another hashing scheme based on ring partition and invariant vector distance [31]. The generated hash sequence incorporates statistical measures such as the mean and variance of each circular ring, which exhibits strong robustness.

Shen et al. [32] utilized the characteristic sensitivity of the human visual cortex to **Color Opponent Components (COCs)**, such as red/green, green/red, blue/yellow, and yellow/blue, to extract color features from images using COC. They also employed **Quadtree Decomposition (QD)** for extracting texture features. QD is an important image representation method that recursively divides the 2D space into a tree-like structure at different levels. When the structure threshold within a region exceeds the difference between the maximum and minimum pixel values, the subdivision of that region stops. The number of block regions used to represent the structural characteristics of the image was determined by the authors. Setyawan et al. [33] employed a histogram of gradients to extract image features. The histogram statistical features were obtained based on the descriptors of oriented gradients. By pseudo-randomly rearranging these statistical features using a key, the final hash sequence can be constructed.

2.1.4 Dimensionality Reduction-Based Schemes. Dimensionality reduction techniques include **Singular Value Decomposition (SVD)**, **Non-Negative Matrix Factorization (NMF)**, **Locally Linear Embedding (LLE)**, PCA, **Compressive Sensing (CS)**, quaternion Zernike moments, and quaternion discrete orthogonal moments [34]. These techniques embedded features of high-dimensional images into low-dimensional spaces, which were able to capture characteristics of digital images successfully under most image attacks [35]. Figure 6 illustrates the development of image hashing schemes based on dimensionality reduction techniques.

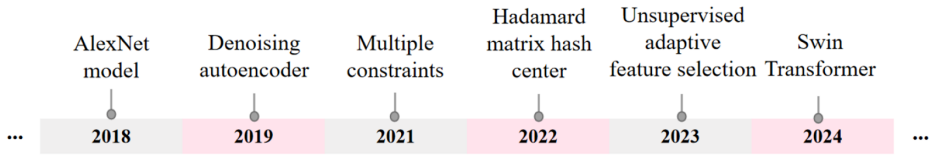


Fig. 7. Development of learning-based perceptual image hashing schemes.

Kozat et al. [36] applied the SVD technique to hashing scheme. Original images were normalized and pseudo-randomly selected overlapping blocks of the same size were chosen. For each image, a smooth quadratic image was constructed using SVD. The process was then repeated by pseudo-randomly selecting overlapping blocks of the same size from the quadratic images and applying the SVD technique to each block. Coefficients from the low-rank approximation matrix were used as the final hash value. In [19], the SVD technique was employed for image pre-processing, followed by perceptual feature extraction based on block truncation coding. Finally, the PCA dimensionality reduction technique was used to generate the final hash sequence. Abbas et al. [37] extracted a **Noise-Resistant Local Binary Pattern (NRLBP)** from non-overlapping blocks of grayscale images. The combination of NRLBP and SVD provided more robustness features. The scheme [38] divided the image into ring partitions and constructed secondary images from these circular regions.

LLE is a non-linear dimensionality reduction scheme that captures the non-linear structure in high-dimensional data by utilizing the local symmetry of linear reconstructions. LLE consists of three steps: neighbor selection, weight computation, and computation of low-dimensional embedding vectors. Tang et al. [39] proposed a perceptual image hashing scheme based on LLE. In their work, the L component of an image was divided into equally sized non-overlapping image blocks. The average pixel values within each block were used to construct a feature matrix. In this feature matrix, key pseudo-randomly selected overlapping blocks of the same size. Pixel values within each block were reorganized from left to right in column direction to form high-dimensional vectors. Afterwards, the LLE technique was applied to this squared image to obtain the final hash.

CS, which leverages advantages of random projection and computational security in sparse signal representation, has been successfully applied to image authentication [40]. The scheme proposed in [40] utilizes CS and the **Fourier-Mellin Transform (FMT)** to extract perceptual features from images. FMT exhibited robustness against attacks such as rotation and scaling, while the inherent dimensionality reduction property of CS made it well-suited for hash function design. Therefore, the scheme in [40] demonstrated satisfactory robustness against geometric distortion in image attacks.

2.2 Learning-Based Perceptual Image Hashing Schemes

Learning-based perceptual image hashing schemes utilized neural network models to automatically learn feature representations of images [78, 79]. Existing schemes can be broadly categorized into two main types, i.e., supervised and unsupervised perceptual image hashing schemes. Figure 7 illustrates the development of learning-based perceptual image hashing schemes.

2.2.1 Supervised Perceptual Image Hashing Schemes. Li et al. [41] proposed a data-driven image hashing scheme, utilizing a neural network model to automatically discover the optimal mapping relationship from images to hash sequences. They employed a **Stacked Denoising Autoencoder (SDAE)**, constructing a five-layer neural network model, where the output of the last layer was used to quantize and generate the final hash sequence. The scheme employed a layer-wise pre-training

and joint training strategy to optimize the model. In the layer-wise pre-training phase, an original image and a transformed identical image were used as input to explore the encoding relationship between the original and perceptual identical images. In the joint training phase, a labeled dataset was constructed, including pairs of both identical and distinct images, for training the SDAE.

Qin et al. [42] proposed an end-to-end perceptual image hashing model under multiple constraints, utilizing a **Convolutional Neural Network (CNN)** for automatic feature extraction to generate the final hash sequence. By combining convolutional layers and pooling layers, the size of input images was reduced while increasing the channel depth for feature extraction. In this scheme, images were divided into original images, perceptual identical images, and perceptual distinct images, which were proportionally inputted into the model as pairs of identical image pairs and distinct image pairs for training. Constraints on image pairs were applied to reduce the hash distance between identical images. The paper designed a strategy to optimize the model by searching for the optimal weighting factors, dynamically optimizing the two sets of symmetric constraints, and achieving a balance between perceptual robustness and discrimination.

Paul et al. [43] proposed a novel image authentication scheme using **Denoising Autoencoders (DAEs)** for encoding images. Additionally, they introduced a new image correction method based on analysis of intrinsic geometric features, allowing the scheme to be robust against compound attacks. By training the model with content-preserving manipulations as noise, input images were mapped to low-dimensional hash sequences through hierarchical encoding. The decoder was then used to generate the hash sequence. Li et al. [44] employed short- and long-range structural dependencies of images to explore the optimal mapping relationship from images to hash sequences, respectively. The Transformer networks were presented to generate hash sequences in [45].

Sun et al. [46] presented a novel perceptual image hashing scheme based on CNN. Each original image along with all its perceptual identical images generated through content-preserving manipulations was treated as a category. Different categories formed distinct pairs for training the hash model using a classification approach, significantly reducing the number of original images required during training. In order to generate more effective and high-quality hash sequences, the concept of hash center was introduced. With the help of center quantization and different quantizations, the similarity relationships between training images were considered globally to constrain distances between their corresponding hash sequences.

Previous learning-based perceptual image hashing schemes all adopted manually designed neural networks. Zhou et al. [47] proposed a perceptual image hashing scheme based on **Neural Architecture Search (NAS)**. The advantage of NAS is that it includes manually designed networks in the search space, and the architecture obtained by NAS is the best result in the search space, thereby surpassing the manually designed networks. For a given image dataset and content-preserving manipulations, based on a continuous search space and considering both the accuracy and computational complexity of the perceptual image hashing task, the corresponding neural network architecture can be automatically obtained.

2.2.2 Unsupervised Perceptual Image Hashing Schemes. Ma et al. [48] proposed an unsupervised perceptual image hashing scheme based on bidirectional generative adversarial networks, which can generate image hash sequences with strong image semantic feature representation capabilities. Additionally, the ResNet18 and gray-level co-occurrence matrix [80] had also been used in the hash generation process. Du et al. [49] proposed a perceptual saliency-based semi-supervised image hashing scheme with multi-view embedding. This scheme incorporated both supervised information and multiple features into the process of training. Additionally, image forgery often involved combinations of images that were highly correlated with human perception, including object addition, deletion, and substitution, which typically affected the regions of interest in related

images. In this scheme, they referred to variations in saliency maps between trusted images and test images as perceptual saliency, considering it as an indication of tampering. The scheme did not learn metric distances on global images but explored local hash distances by considering the perceptual saliency effected between different regions. However, the multi-view features were only used in combination and not learned through feature learning.

Shaik et al. [50] utilized a **Convolutional Stacked DAE (CSDAE)** to generate robust hash sequences for different content-preserving manipulations. The CSDAE scheme mapped high-dimensional data information to low-dimensional hash sequences while preserving semantic similarity, ensuring that images with similar content have similar hash sequences. The scheme followed an encoder-decoder architecture, where noisy images were input to the encoder and the corresponding denoised images served as the output to train the model. Noisy images subjected to various manipulations such as changes in brightness, contrast, gamma correction, adding Gaussian noise, salt and pepper noise, speckle noise, image scaling, rotation, and compression. Denoised images represented the reconstructed images generated by the decoder from the output of the encoder. By continuously optimizing the L_2 distance between denoised and original images, the encoder became robust to content-preserving manipulations.

Jesus et al. [51] investigated the application of self-supervised learning in digital image authentication hashing and developed a deep image content authentication scheme based on contrastive learning called ProtoHash. The scheme initially trained the ProtoHash model utilizing an unlabeled image database through self-supervised learning, employing a proxy task. A set of images extracted from the dataset was pre-processed using data augmentation operations to present the model with perceptual information under various attacks. Subsequently, the feature representation of the input image was obtained from the last fully connected layer of the trained ProtoHash model. Finally, the binary value of the image was generated by quantizing the feature representation with a threshold.

3 Perceptual Video Hashing

Video is also an important type of multimedia data, and researchers have designed perceptual hashing techniques for video content [81]. The perceptual features of video usually contain features in both temporal and spatial domains. The temporal domain features describe the changing process of video over time and reflect the dynamic changing scene information. The spatial domain features capture the visual content at a single frame and present a static image. According to the different ways of extracting image features, perceptual video hashing schemes can be divided into classical methods-based schemes and learning-based schemes. Classical methods-based schemes are mainly divided into spatial feature-based schemes and spatial-temporal feature-based schemes. Learning-based schemes can be divided into supervised and unsupervised perceptual video hashing schemes.

3.1 Classical Methods-Based Perceptual Video Hashing Schemes

3.1.1 Spatial Feature-Based Schemes. Naturally, early perceptual video hashing schemes focused on spatial features, they simply divided the video into frames and applied image hashing schemes to keyframes, and then performed feature statistics on the obtained frames to generate the final video hash sequence. Figure 8 illustrates the development of classical methods-based perceptual video hashing schemes.

Neelima et al. [52] used SIFT technology to extract and cluster invariant keypoints in video keyframes, and then constructed a matrix of 32 keypoints from all clustered centers and extracted the maximum singular values as feature vectors by SVD to generate the final video hash sequence. Yang et al. [53] extracted **Speeded Up Robust Features (SURF)** points at the frame level by dividing the frame into 16 equal non-overlapping square blocks, and then traversed the number of SURF points in each quadrant to construct a third-order Hilbert curve across each quadrant, and generated an

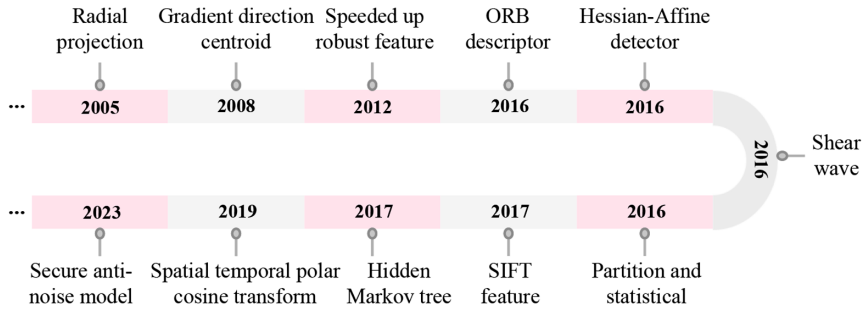


Fig. 8. Development of classical methods-based perceptual video hashing schemes.

adjacent grid that maintains the same neighborhood relationship with the original frame. Finally, the hash sequence was generated by calculating the differences of the SURF points. Sun et al. [54] designed a perceptual video hashing scheme based on **Hidden Markov Tree (HMT)**. First, video frames were normalized and divided into blocks. Secondly, the contour wave transformation was applied to each block to obtain the contour wave coefficients. Then the coefficients were inputted into the HMT model to generate the standard deviation matrix, and the dimensionality was reduced by the SVD method. Finally, the intermediate fingerprint segments extracted from all the frames are concatenated to output the video hash sequence.

Lee and Yoo [55] proposed a robust video hashing scheme based on the gradient direction centroid. First, the input video was resampled at a fixed frame rate. Secondly, resampled frames were converted to grayscale, which made the proposed scheme robust to color change. Then grayscale frames were resized to normalize their width and height. Finally, these frames were divided into a grid of rows and columns to form blocks, and the gradient direction centroid was further calculated for each block to obtain the hash vector of each frame. Roover et al. [56] proposed a robust hashing scheme (**Radial Hashing (RASH)**) for static images based on radial projection of image pixels. They summarized the video content into keyframes and used a RASH vector to characterize each keyframe. In particular, they focused on the projection of image pixels on a line bundle that passed through the center of the image and performed feature extraction based on its angular orientation. For each projection angle, the feature sample was defined as the variance of pixel brightness value on the corresponding projection line. This scheme captured the brightness discontinuity along the projection line to extract the image edge distribution information, so as to effectively represent the visual content of an image. Compared with the histogram-based feature vectors, the proposed RASH feature vector exhibited higher robustness and uniqueness.

Mao et al. [57] assumed that the probability of five consecutive identical scene frames appearing in two different videos was very low. First, five consecutive frames were extracted from the video, scaled, and divided into subregions. Then, the mean value and another four different elements were extracted from the brightness component of the subregion. This process generated 144 average values and 576 difference values, and finally the intermediate features were quantized and clustered to output the final hash sequence.

Chen et al. [58] proposed a novel robust video hashing scheme based on the **Secure Anti-Noise Model (SANM)**. To improve the robustness against noise attacks, the input video was reconstructed into SANM by Low-Rank Representation and Random Subspace Partition. In addition, the scheme proposed a new **Subspace Decomposition Descriptor (SDD)**. The SDD was obtained by computing the invariant distance of the factor matrix obtained by the Tucker decomposition, and it is used to decompose the SANM to generate compact video hash sequences.

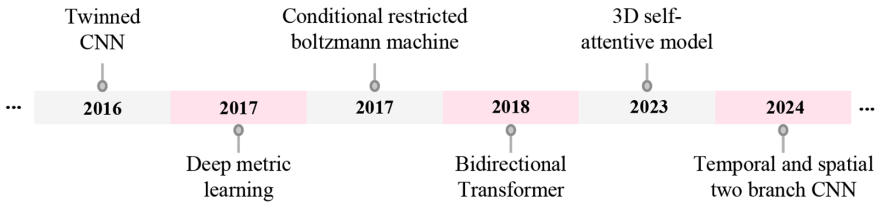


Fig. 9. Development of learning-based perceptual video hashing schemes.

Garboan et al. [59] considered the affine covariance region based on the Hessian-Affine detector as a localized feature and further computed the SIFT descriptors on the Hessian-Affine region transformed into a circular shape. Then, SIFT descriptors were clustered into a smaller set. Finally, the Bag of Words histogram of the frame was calculated and used as visual word frequency vectors, and components of such vectors were weighted using the Term frequency-inverse Document Frequency weighting scheme [82] to generate the video hash sequence.

3.1.2 Spatial and Temporal Features-Based Schemes. Subsequent perceptual video hashing schemes combine temporal and spatial features to jointly generate a robust video hash sequence.

Yuan et al. [60] proposed a shear wave-based video hashing scheme, which took both low- and high-frequency coefficients into account in the process of hash generation. Among them, the low-frequency coefficients focused on the large-scale structure in the visual content, which were stable to the common video content-preservation manipulations and provided the robustness of the scheme. The high-frequency coefficients captured the detailed features of visual content, such as the edges of the image, texture details, and so forth, which were used to enhance the discrimination capability of the scheme. Tang et al. [61] proposed a geometrically robust video hashing scheme for video copy detection based on Spatial-Temporal **Polar Cosine Transform (PCT)**. They performed 2D PCT on the video spatial domain and 1D DCT on the video temporal domain. The scheme treated the video as a 3D matrix and extracted the video hash sequence based on geometric invariance. Ren et al. [62] selected keyframes by calculating the histogram-based contrast as well as the underlying image entropy and combined it with Pearson's correlation coefficient. They extracted the global and local spatial features of the keyframes by using the color histogram and the ORB descriptor, respectively. Next, the local temporal domain information was captured by calculating the angular histograms of the motion vectors. Therefore, the keyframe feature was a combination of color histogram, ORB descriptor, and motion vector normalized histogram, and the final video hash sequence was a collection of keyframe features.

3.2 Learning-Based Perceptual Video Hashing Schemes

The breakthrough of deep learning technology provides a new direction for the study of perceptual video hashing. Researchers use the powerful feature learning and quantification capabilities of deep learning models to automatically extract perceptual features of the video and map them into hash sequences. The existing learning-based schemes can be divided into two main categories: supervised perceptual video hashing schemes and unsupervised perceptual video hashing schemes. Figure 9 illustrates the development of learning-based perceptual video hashing schemes.

3.2.1 Supervised Perceptual Video Hashing Schemes. Jiang et al. [63] first proposed a dataset for evaluating video copy detection schemes and then proposed a perceptual video hashing framework called twinned CNNs. They considered two video images of the same video, or one original video image and one video image after content-preserving manipulations as positive sample pairs, and the remaining combination of image frames as negative sample pairs. They used two identical network

architectures to extract two frames of video image features and calculated the Euclidean distance between the two frames of image features. By contrast loss function, the distance between the positive sample pairs is shortened, and the distance between the negative sample pairs is pushed away, so as to generate a robust video hash sequence. Giorgos et al. [64] proposed a perceptual video hashing network architecture based on deep metric learning. They first used a pre-trained CNN to extract video image features and then converted features extracted from each convolutional layer in the neural network into a single descriptor through the maximum pooling operation. The author referred to as max-activation of the convolution, and finally splicing the descriptors of all layer vectors into a hash sequence after normalization. The scheme also used triplet network input that divided the training datasets into original, perceptual identical, and perceptual distinct videos.

Li et al. [65] adopted a **Conditional Restricted Boltzmann Machine (CRBM)** [83] to simultaneously extract the spatial and temporal information of video. In addition, the scheme also combined CRBM with a DAE to discover the invariants of the video distortion content-preservation manipulation, which reduced the dimension of the CRBM model output and the redundancy of the hash sequence. Xu et al. [66] proposed a two-branch deep perceptual video hashing network. The two branches of the network focused on the temporal and spatial information of the video, respectively. The temporal branch consisted of a temporal residual bottleneck structure for capturing subtle motion variations and a temporal attention module for obtaining the overall motion trend, so as to comprehensively capture the complex motion pattern of the video subject. The spatial branch utilized several different receptive fields to extract multi-scale spatial information and retained the structural features of each frame. Deep metric learning and hard triplet loss function are used to constrain the generation of video hash sequences.

3.2.2 Unsupervised Perceptual Video Hashing Schemes. Song et al. [67] proposed a self-supervised perceptual video hashing scheme based on a bidirectional Transformer [84], which was originally used in the field of video cation recognition tasks. Due to its powerful feature extraction capability, this scheme can also be used to extract robust video features to generate hash sequences. Using Transformer's encoder-decoder structure, a visual completion task was designed to fully capture the bidirectional correlation between video frames, and a self-supervised learning framework was designed to learn unique binary hash sequences. Zhao et al. [68] proposed an unsupervised scheme for video copy detection. The scheme proposed a 3D self-attentive multi-scale feature fusion model based on which the spatial-temporal information in the video can be effectively fused and condensed into high-dimensional features. The perceptual content of the video is identified by calculating the correlation distances between these features to generate pseudo-labels for the video, which in turn guided the network training and generated robust hash sequences. In addition, the dual constraint is introduced to make the generated hash sequence more robust.

4 Perceptual Hashing for Neural Network Model

In recent years, neural network technology has made breakthroughs in different fields, and many research institutions and enterprises have invested a lot of manpower and material resources in the research of neural network models. However, the problem of model piracy is becoming more and more serious. Considering the high value of the models, potential pirates may steal the models for illegal profits without the permission of the holders, which violates the intellectual property rights of the model designer. Therefore, in recent years, the protection of neural network models has also gradually become a worth studying topic. The input of perceptual model hashing is usually the parameters of the model or other main features of the model, emphasizing the security of the protected model.

Chen et al. [69] converted the parameter weights of the neural network model into a fixed-length binary hash sequence so that the slightly modified model has a hash sequence similar to that of the original model. By comparing the similarity of the hash sequence between the models in the copyright protection library and the model to be tested, the similar version of the model can be effectively queried to determine whether the model under test is infringing. They first extracted the important parameters of the model based on the model compression theory and then calculated the data distribution of these parameters as the hash sequence of the model. Xiong et al. [70] used neural networks to extract model parameter features and convert them into hash sequences. Considering the large number and wide range of model parameters, they first pre-processed the model parameters and divided the model parameters into convolution layer parameters and fully connected layer parameters. Then, the number of model parameters is reduced by pruning, and the size of model parameter matrix of each layer is adjusted by the adaptive mean pooling method, so that the size of the model parameter matrix of different network layers is adjusted to the same size. The numerical range of model parameters is adjusted by the local normalization method. A multi-layer convolutional layer is used to extract features, and dimensionality reduction and compression of the features are realized by optimal value pooling. Finally, the fully connected layer converted the feature into a hash sequence.

Inspired by the transferability of adversarial examples, Zhao et al. [71] proposed a neural network model hashing scheme by Adversarial-Marks. They found that the adversarial examples not only fool the attacked model, but can also be transferred to other models. They therefore strengthened this transferability to make the Adversarial-Marks robust to simple model changes, so that they could be better transferred to the plagiarism models rather than to other irrelevant models. Since the plagiarists still want to maintain the model performance, the small model modifications they make are not enough to change or remove these inherent fingerprints of the original model. Then they can detect these fingerprints in the plagiarism models to realize the authentication of neural network models. Zhu et al. [72] proposed a supervised two-stage perceptual hashing scheme for neural network models based on graph neural networks. In the first stage, they used a graph generator to convert the neural network model to an undirected graph and extracted additional information after a series of content-preserving manipulations. Then they used the graph hashing network to generate compact hash sequences in the second stage.

5 Performance Analysis

In this section, we present the datasets used for the perceptual hashing experiments, and the main evaluation metrics used to evaluate perceptual hashing schemes, including perceptual robustness, discrimination, key-dependent security, and complexity. Since the number of perceptual image hashing schemes far exceeds those in other objects (videos or neural network models), and the perceptual hashing schemes of various objects are similar in terms of evaluation metrics, this section takes perceptual image hashing as an example to introduce.

5.1 Datasets

5.1.1 Datasets and Existing Problems. Datasets for perceptual image hashing schemes generally consist of three parts: original images, perceptual identical images, and perceptual distinct images. Previous datasets usually select original images from datasets in other image processing datasets, such as COCO dataset [85], CASIA dataset [86], ImageNet dataset [87], RAISE dataset [88], UCID dataset [89], USC-SIPI dataset [90], and Pascal VOC 2012 dataset [91]. The designed content-preserving manipulations are then applied to these original images to generate perceptual identical images, and then together with randomly selected perceptual distinct images to form the perceptual image hashing dataset.

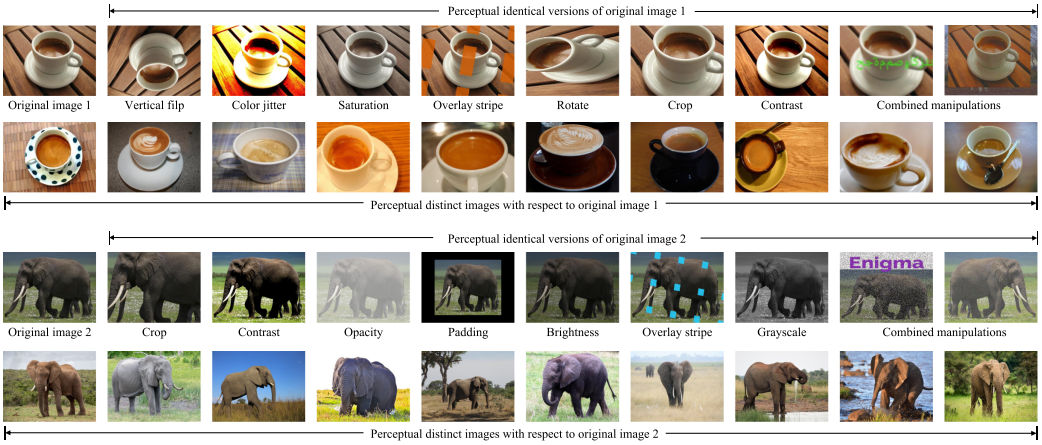


Fig. 10. Image examples in the dataset proposed by Zhou et al. [47].

These datasets often suffer from the following problems. First of all, the types of content-preservation manipulations used in these datasets are generally no more than 10, and these content-preserving manipulations were relatively simple, which only include common manipulations such as cropping, rotation, and noise. Such manipulations could no longer cope with the increasingly complex image authentication scenarios. The second problem is that in the previous datasets, the perceptual distinct images are often randomly selected, resulting in a large difference between the original image and the perceptual distinct images, which enables the perceptual image hashing schemes to easily learn the difference between the features of these images, so as to achieve good authentication performance on these datasets. Therefore, the performance of these schemes degrades significantly when faced with more complex authentication scenarios in real-world.

5.1.2 Dataset for Perceptual Image Hashing Task. Zhou et al. [47] summarized various problems of previous perceptual image hashing datasets and proposed a challenging dataset for perceptual image hashing tasks. Their dataset is also divided into three parts, namely original, perceptual identical, and perceptual distinct images. Figure 10 shows some image examples in their dataset.

Original images of their dataset are widely sourced. They first selected original images from ImageNet dataset [87]. There are 1,000 classes of image data in this dataset, and the classes of images involved are also very rich, such as animals, plants, human movement, and landscapes. They chose one image for each class as the original image. In addition to selecting from the existing dataset, they also selected images from the Internet as original images. They randomly generated names of objects and then selected corresponding images from the Internet.

For perceptual identical images, they generally divided the content-preserving manipulations into four categories: geometric manipulation (including image scaling, rotation, cropping, etc.), enhancement manipulation (including contrast adjustment, brightness adjustment, blur, etc.), filter manipulation (including different kinds of image filters), and editing manipulation (adding stripe or text). For each manipulation, they randomly set its parameters in a limited range. In real applications, image editing is often not a single manipulation, so they also designed combined manipulations, i.e., each of them is obtained by randomly selecting four kinds of single manipulations. The type, number, and range of parameters for the manipulations are listed in Table 2. Figure 10 shows two examples of original images in their dataset, along with some corresponding perceptual identical and distinct images. The first and third rows show some examples of the perceptual identical images,

Table 2. The Content-Preserving Manipulations and Parameter Settings of the Image Dataset Proposed by Zhou et al. [47]

Types	Number	Manipulations	Parameters
Geometric	1	Flip	direction ∈ horizontal (H)
	2	Flip	direction ∈ vertical (V)
	3	Rotate	degree ∈ (60°, 300°)
	4	Crop	scale ∈ (0.64, 0.81)
	5	Scale	scale ∈ (0.5, 3)
	6	Resize	width, height ∈ (200, 500)
	7	Clip	scale ∈ (200 × 200, 1,200 × 1,200)
	8	Aspect ratio	scale ∈ (0.5, 2)
Enhancement	9	Blur	scale ∈ (1.5, 2.5)
	10	Brightness	scale ∈ (0.5, 1.5)
	11	Contrast	scale ∈ (0.5, 2)
	12	Color jitter	mode = {brightness, contrast, saturation}
	13	Encoding	scale ∈ (5, 50)
	14	Grayscale	mode = {luminosity, average}
	15	Opacity	scale ∈ (0.4, 0.8)
	16	Pixelization	scale ∈ (0.2, 0.5)
	17	Saturation	scale ∈ (0.3, 3)
	18	Sharpen	scale ∈ (0.5, 5)
	19	Shuffle pixel	scale ∈ (0.2, 0.4)
	20	Random noise	mean ∈ (0.2, 0.5), var ∈ (0.01, 0.1)
	21	Masking	mask = {1, L, RGBA}
	22	Padding	pad color = random, pad size ∈ (0.05, 0.2)
Filter	23	Detail filter	scale ∈ (0.5, 3)
	24	Edge filter	scale ∈ (0.5, 3)
	25	Smooth filter	scale ∈ (0.5, 3)
	26	Kernel filter	scale ∈ (0.5, 3)
	27	Rank filter	scale ∈ (0.5, 3)
	28	Max filter	scale ∈ (0.5, 3)
	29	Min filter	scale ∈ (0.5, 3)
	30	Median filter	scale ∈ (0.5, 3)
	31	Mode filter	scale ∈ (0.5, 3)
Editing	32	Background	size ∈ (0.8, 1.2)
	33	Overlay stripe	color, direction = random, line width ∈ (0.1, 0.4)
	34	Overlay text	text size ∈ (0.05, 0.2)
	35	Meme	text = {“AugLy,” “Meta,” “Apple”}
Combined		Combination	Randomly combine four kinds of manipulations

which are obtained from original images after conducting corresponding single and combined content-preserving manipulations.

For each original image, perceptual distinct images are selected from other images with the same class. The second and fourth rows of Figure 10 show some examples of perceptual distinct images with respect to the corresponding original images. This design is to better fit the perceptual image hashing task because perceptual image hashing is to authenticate the image content and does not involve the image semantic information. In other words, images of the same class should also be regarded as perceptual distinct.

5.2 Perceptual Robustness

In the Introduction section, we discuss the concept of perceptual robustness, which emphasizes that two perceptual identical images should produce the same or extremely close hash sequences [92]. Perceptual image hashing schemes either select test images from standard datasets or create their own datasets. Content-preserving manipulations are subsequently applied to generate perceptual identical images corresponding to the test images.

To validate the perceptual robustness of the hashing schemes, we tested the attack robustness of 1,338 images from the UCID dataset [89] under 10 content-preserving manipulations. These manipulations include JPEG compression, gamma correction, Gaussian noise, speckle noise, salt and pepper noise, circular blur, median filtering, Gaussian filtering, rotation, and scaling. Through these 10 content-preserving manipulations, a total of 100 perceptual identical images were generated for each original image, resulting in a total of 135,138 images (1,338 × 100 + 1,338). The hash distances between the hash sequences of each original image and its perceptual identical images were calculated. We compared the performance of 10 representative perceptual image hashing schemes. These schemes included local feature-based schemes [19, 23], transform-based schemes [25, 28], statistical feature-based schemes [30, 31], dimensionality reduction-based schemes [34, 38], and learning-based schemes [41, 42]. As shown in Table 3, all these schemes demonstrate good performance in terms of perceptual robustness. Hash distances of schemes [19, 23, 25, 30, 31, 34, 42] measured by normalized Hamming distance and Euclidean distance are sufficiently small, while schemes [28, 38, 41] measured by correlation coefficient are close to 1.

5.3 Discrimination

Discrimination capability is another crucial property of perceptual image hashing schemes, where two perceptual distinct images should produce significantly different hash sequences. We also tested the discrimination capability of 10 representative hashing schemes [19, 23, 25, 28, 30, 31, 34, 38, 41, 42] mentioned above. We generated hash sequences between 1,338 uncompressed color images from the UCID dataset [89] and plotted the distribution of hash distances (such as correlation coefficient, Euclidean distance, and normalized Hamming distance). The abscissa of Figure 11 represents the hash distance, and the ordinate represents the frequency of the corresponding hash distance. Each subfigure contained 894,453 hash distances, corresponding to $C^2_{1,338}$ pairs. These data were then organized into a table to calculate the probability of hash sequence collisions under each threshold, and the results were listed in Table 4. Except for schemes using correlation coefficient as distance measurement, lower thresholds always result in lower collision probabilities, but may compromise the perceptual robustness. An appropriate threshold should be chosen based on the specific application to achieve an ideal tradeoff between perceptual robustness and discrimination capability [93].

As listed in Table 4, the learning-based scheme [42] demonstrates the best discrimination capabilities, with a collision probability of 0 for authentication thresholds ranging from 0.1 to 0.3. At a higher threshold of 0.4, the collision probability increases to 1.1163×10^{-6}. Schemes [41] and [28] both take the correlation coefficient as the distance measurement, and the collision probability gradually decreases when the authentication threshold gradually approaches 1. These two schemes demonstrate moderate discrimination capabilities. For scheme [41], setting the authentication threshold at 0.96 yields a collision probability of 4.4653×10^{-6}, whereas at 0.98, it reduces to 1.1163×10^{-6}. Similarly, the collision probability of scheme [28] is reduced to the lowest 1.1163×10^{-6}, when the threshold is 0.96. The threshold of schemes [31] and [34] vary widely, and although the collision probability seems to be large from Table 4, they still perform good discrimination capabilities. Scheme [19] demonstrates slightly worse discrimination capabilities than the previous

Table 3. The Hash Distance Statistics of Schemes under Different Content-Preserving Manipulations

Scheme	Content-preserving manipulation	Perceptual robustness hash distances				Scheme	Content-preserving manipulation	Perceptual robustness hash distances			
		Min.	Max.	Mean	Std.			Min.	Max.	Mean	Std.
	JPEG compression	0.0004	4.0129	0.03807	0.0154		JPEG compression	0	0.1359	0.0082	0.0001
	Gaussian noise	0.0015	1.9023	0.07632	0.06452		Gaussian noise	0.0047	0.4875	0.1856	0.0075
	Speckle noise	0.0001	1.0560	0.1206	0.0093		Speckle noise	0	0.4734	0.1162	0.0049
	Salt and pepper noise	0.0009	0.9730	0.1829	0.1409		Salt and pepper noise	0.0062	0.4500	0.1831	0.0081
	Circular filtering	0.0006	1.3229	0.5012	0.4529		Circular filtering	0.0015	0.4468	0.2090	0.0071
SSF [23]	Median filtering	0.0001	0.7289	0.0049	0.0028	BTC [19]	Median filtering	0	0.3468	0.0411	0.0013
	Gaussian filtering	0.0001	0.8022	0.0032	0.0018		Gaussian filtering	0	0.2312	0.0113	0.0001
	Gamma correction	0.0002	0.4217	0.0071	0.0025		Gamma correction	0.0312	0.7609	0.3426	0.0159
	Rotation	0.0003	0.7802	0.0903	0.0793		Rotation	0.0070	0.4296	0.2138	0.0072
	Scaling	0.0001	0.5792	0.0412	0.0018		Scaling	0	0.3515	0.0143	0.0001
	JPEG compression	5.0947	1.1887	2.5560	1.3213		JPEG compression	−0.9707	0.9999	0.9896	0.0089
	Gaussian noise	0.3673	0	0.1787	0.1140		Gaussian noise	−0.9931	0.9999	0.9561	0.0383
	Speckle noise	6.9147	0.7813	2.7980	1.5900		Speckle noise	−0.9961	0.9999	0.9602	0.0442
	Salt and pepper noise	9.2440	1.4013	3.9607	2.0080		Salt and pepper noise	−0.9951	0.9999	0.9589	0.0381
	Circular filtering	4.5093	1.0092	2.0450	1.2265		Circular filtering	−0.9896	0.9999	0.8683	0.1153
NDW [25]	Median filtering	5.5243	0.2204	2.5664	1.4423	DWT [28]	Median filtering	−0.9715	0.9999	0.9792	0.0199
	Gaussian filtering	6.1293	0.9952	2.4693	1.6632		Gaussian filtering	−0.9624	1	0.9981	0.0022
	Gamma correction	5.2973	0.7707	2.3453	1.3334		Gamma correction	−0.9656	0.9999	0.9801	0.0146
	Rotation	6.7413	1.1153	2.6833	1.4180		Rotation	−0.9808	0.9998	0.7529	0.1284
	Scaling	5.1040	0.9980	2.4200	1.3693		Scaling	0.4021	1	0.9914	0.0004
	JPEG compression	0.0004	2.7762	0.2021	0.2055		JPEG compression	0	0.4863	0.0362	0.0178
	Gaussian noise	0.0003	1.8265	0.0348	0.0399		Gaussian noise	0.0153	0.9605	0.2237	0.1409
	Speckle noise	0.0022	0.7901	0.0578	0.0607		Speckle noise	0.0041	1.0132	0.1192	0.0947
	Salt and pepper noise	0.0006	1.6766	0.1029	0.1034		Salt and pepper noise	0.8832	0.9745	0.1713	0.1331
	Circular filtering	0.0001	0.4876	0.0037	0.0046		Circular filtering	0.5916	1.1218	0.4269	0.4478
RBE [30]	Median filtering	0.0001	0.5668	0.0087	0.0102	IVD [31]	Median filtering	0	7.2445	0.07546	0.0610
	Gaussian filtering	0.0002	0.9812	0.0116	0.0127		Gaussian filtering	0	0.7985	0.0309	0.01653
	Gamma correction	0.0001	0.0213	0.0003	0.0004		Gamma correction	0.0002	0.4389	0.06338	0.02840
	Rotation	0.0036	2.3676	0.3077	0.3589		Rotation	0.0033	1.1320	0.1641	0.1392
	Scaling	0.0043	1.8922	0.2433	0.2677		Scaling	0.0014	0.5841	0.0418	0.0131
	JPEG compression	0.5201	0.9999	0.9998	0.2224		JPEG compression	0.0004	1.1376	0.0093	0.0275
	Gaussian noise	0.3933	0.9999	0.9728	0.0013		Gaussian noise	0.0044	2.6181	0.2411	0.2508
	Speckle noise	0.3445	0.9999	0.9818	0.0013		Speckle noise	0.0031	2.6501	0.1349	0.1741
	Salt and pepper noise	0.1868	0.9999	0.9812	0.0022		Salt and pepper noise	0.0003	0.6708	0.0121	0.0182
	Circular filtering	0.0971	0.9998	0.9538	0.0046		Circular filtering	0.0042	2.6394	0.2191	0.2294
NMF [38]	Median filtering	0.2251	0.9999	0.9956	0.0005	CCA [34]	Median filtering	0	1.9668	0.0393	0.0861
	Gaussian filtering	0.2362	1	0.9973	0.0007		Gaussian filtering	0	0.0212	0.0012	0.0012
	Gamma correction	0.2606	0.9999	0.9923	0.0006		Gamma correction	0.0049	4.0185	0.3059	0.3099
	Rotation	0.2855	0.9999	0.9988	0.0002		Rotation	0.0019	1.8772	0.1036	0.1319
	Scaling	0.2372	0.9999	0.9971	0.0009		Scaling	0.0007	1.5935	0.0118	0.0239
	JPEG compression	0.9737	0.9999	0.9996	0.0101		JPEG compression	0	0.9699	0.0045	0.0223
	Gaussian noise	0.9558	0.9999	0.9961	0.0855		Gaussian noise	0.0002	0.5421	0.02023	0.0224
	Speckle noise	0.9737	0.9999	0.9996	0.0101		Speckle noise	0.0002	0.5768	0.0167	0.0315
	Salt and pepper noise	0.9771	0.9999	0.9982	0.0335		Salt and pepper noise	0.0003	0.6708	0.0121	0.0182
	Circular filtering	0.7414	0.9999	0.9734	0.0005		Circular filtering	0.0001	0.5814	0.0246	0.0358
DAE [41]	Median filtering	0.9512	1	0.9978	0.0145	CNN [42]	Median filtering	0	1.0525	0.0141	0.0399
	Gaussian filtering	0.9999	1	0.9999	0.0158		Gaussian filtering	0	0.0016	0.0003	0.0008
	Gamma correction	0.9321	0.9999	0.9921	0.6095		Gamma correction	0.0002	2.4821	0.0601	0.1007
	Rotation	0.5904	0.9999	0.9143	0.0035		Rotation	0.0001	1.9955	0.1127	0.1294
	Scaling	0.9763	0.9999	0.9988	0.0157		Scaling	0	0.0929	0.0011	0.0031

schemes. In particular, at an authentication threshold of 0.2, the collision probability stands at 1.1163×10^{-6}. At a higher threshold of 0.29, the collision probability increases to 1.2056×10^{-4}.

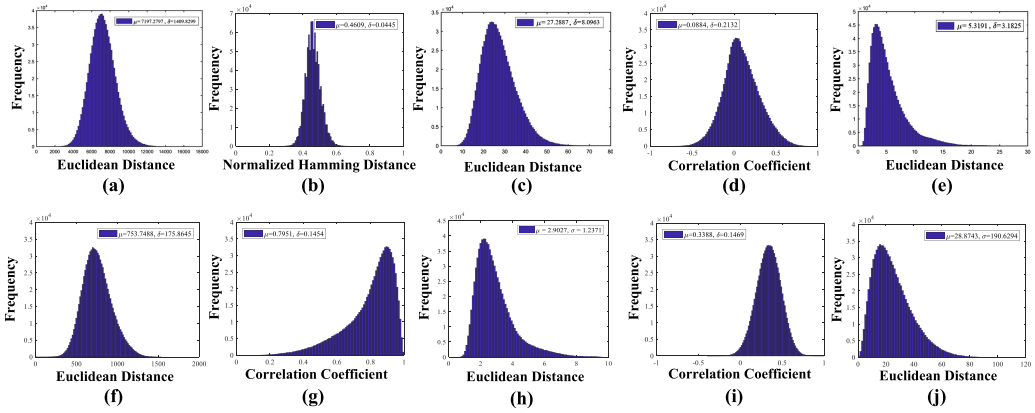


Fig. 11. The histogram of hash distances for different schemes. (a) SSF [23], (b) BTC [19], (c) NDW [25], (d) DWT [28], (e) RBE [30], (f) IVD [31], (g) NMF [38], (h) CCA [34], (i) DAE [41], and (j) CNN [42].

Table 4. Collision Probabilities P_c with Different Thresholds T

Scheme	T	P_c	Scheme	T	P_c	Scheme	T	P_c	Scheme	T	P_c	Scheme	T	P_c
	100	1.1163×10^{-6}		0.20	1.1163×10^{-6}		2.0	1.1163×10^{-6}		0.96	1.1163×10^{-6}		0.30	1.1163×10^{-6}
	120	2.2327×10^{-6}		0.21	2.2327×10^{-6}		2.1	2.2327×10^{-6}		0.95	2.2327×10^{-6}		0.32	3.3490×10^{-6}
	140	5.5817×10^{-6}		0.22	4.4653×10^{-6}		2.2	3.3490×10^{-6}		0.94	3.3490×10^{-6}		0.34	6.6980×10^{-6}
	160	8.9307×10^{-6}		0.23	6.6980×10^{-6}		2.3	5.5817×10^{-6}		0.93	5.5817×10^{-6}		0.36	1.4512×10^{-5}
	180	1.5629×10^{-5}		0.24	8.9307×10^{-6}		2.4	7.8141×10^{-6}		0.92	1.0047×10^{-5}		0.38	2.6792×10^{-5}
SSF [23]	200	2.6792×10^{-5}	BTC [19]	0.25	1.5629×10^{-5}	NDW [25]	2.5	1.3396×10^{-5}	DWT [28]	0.91	1.1163×10^{-5}	RBE [30]	0.40	4.6770×10^{-5}
	220	4.0188×10^{-5}		0.26	2.6792×10^{-5}		2.6	2.6792×10^{-5}		0.90	1.5629×10^{-5}		0.42	7.4794×10^{-5}
	240	6.8096×10^{-5}		0.27	4.6770×10^{-5}		2.7	4.0188×10^{-5}		0.89	1.6745×10^{-5}		0.44	9.3772×10^{-5}
	260	9.3772×10^{-5}		0.28	7.4794×10^{-5}		2.8	5.3584×10^{-5}		0.88	3.3490×10^{-5}		0.46	1.0047×10^{-4}
	280	1.2056×10^{-4}		0.29	1.2056×10^{-4}		2.9	7.4794×10^{-5}		0.87	3.0141×10^{-5}		0.48	3.3490×10^{-4}
	80	1.1163×10^{-6}		0.999	1.1163×10^{-6}		0.50	1.1163×10^{-6}		0.98	1.1163×10^{-6}		0.1	0
	100	3.3490×10^{-6}		0.998	1.3396×10^{-5}		0.55	2.2327×10^{-6}		0.96	4.4653×10^{-6}		0.2	0
	120	5.5817×10^{-6}		0.997	3.3490×10^{-5}		0.60	7.8141×10^{-6}		0.94	6.6980×10^{-6}		0.3	0
	140	8.9307×10^{-6}		0.996	9.3772×10^{-5}		0.65	1.8978×10^{-5}		0.92	1.1163×10^{-5}		0.4	1.1163×10^{-6}
	160	1.4512×10^{-5}		0.995	1.8308×10^{-4}		0.70	3.6838×10^{-5}		0.90	1.8978×10^{-5}		0.5	2.2327×10^{-6}
IVD [31]	180	3.4606×10^{-5}	NMF [38]	0.994	3.3043×10^{-4}	CCA [34]	0.75	8.3723×10^{-5}	DAE [41]	0.88	4.0188×10^{-5}	CNN [42]	0.6	3.3490×10^{-6}
	200	5.3584×10^{-5}		0.993	5.3696×10^{-4}		0.80	1.6745×10^{-4}		0.86	6.8096×10^{-5}		0.7	5.5817×10^{-6}
	220	6.6980×10^{-5}		0.992	7.6479×10^{-4}		0.85	3.3490×10^{-4}		0.84	1.3284×10^{-4}		0.8	1.1163×10^{-5}
	240	1.0270×10^{-4}		0.991	1.0616×10^{-3}		0.90	5.5817×10^{-4}		0.82	2.3220×10^{-4}		0.9	1.7861×10^{-5}
	260	1.4401×10^{-4}		0.990	1.4414×10^{-35}		0.95	9.4886×10^{-4}		0.80	4.0746×10^{-4}		1	2.3443×10^{-5}

Scheme [38] demonstrates relatively poor discrimination capabilities, even in a small threshold range, the collision probability of the scheme is very large.

5.4 Key-Dependent Security

Key-dependent security requires that the generation of image hash sequences is controlled by one secret key. With this property, different secret keys applied to the same image will yield different corresponding hash sequences.

In our security experiments, the distances between hash sequences with correct and wrong keys for each image were calculated. We randomly generated 1,000 wrong keys and extracted the hash sequences of the 1,000 images in COCO dataset [85] with each wrong key, respectively. These 1,000 hash distances are plotted across various subfigures of Figure 12. In each subfigure, the

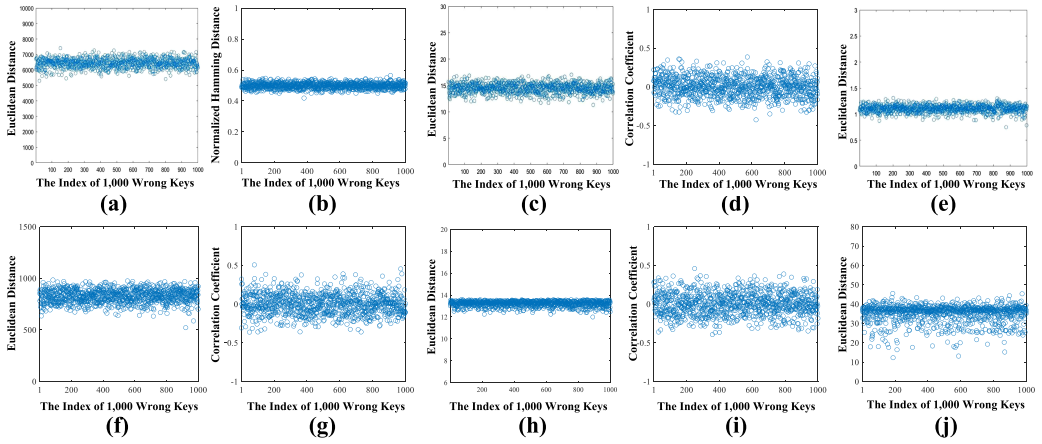


Fig. 12. Results of key-dependent security for different schemes. (a) SSF [23], (b) BTC [19], (c) NDW [25], (d) DWT [28], (e) RBE [30], (f) IVD [31], (g) NMF [38], (h) CCA [34], (i) DAE [41], and (j) CNN [42].

abscissa is the index of 1,000 wrong keys and the ordinate is the hash distance between correct and wrong keys.

In Figure 12(b), the normalized Hamming distances of scheme [19] are tightly clustered around 0.5, demonstrating a compact data distribution. Consequently, the key security of scheme [19] appears to be robust. Moving to Figure 12(i), (d), and (g), the correlation coefficients of schemes [28, 41] and [38] exhibit a general distribution between −0.3 and 0.3, indicating a moderate level of key security for these schemes. Figure 12(a), (c), (e), and (f) depict hash distances in Euclidean distance. The scheme is considered key-dependent if the distance of the hash sequence generated with a wrong key is greater than the threshold for perceptual identical. For example, hash distances of scheme [25] generated by wrong keys are predominantly distributed between 10 and 20, but the threshold of scheme [25] is 3. This means that hash sequences generated by these wrong keys will be judged as perceptual distinct, so this scheme is key-dependent. Hash distances of Figure 12(h) and (j) are distributed in 10 to 50 and 12 to 14, respectively. Schemes [42] and [34] also have satisfactory security performance.

5.5 Complexity

The complexity of the hash scheme consists of two main components, i.e., computational complexity and the length of hash sequence. The computational complexity measures the average time required to generate a hash sequence. The length of hash sequence mainly affects the transmission and storage space requirements.

We conducted experiments on computational complexity and hash length for 10 hashing schemes [19, 23, 25, 28, 30, 31, 34, 38, 41, 42] using 1,338 images from the UCID dataset [89]. Each scheme generated 1,338 hash sequences from the aforementioned images, and the average computation time for generating hash sequences was summarized in the second column of Table 5; all results were obtained on a 3.60 GHz Intel Xeon W-2123 CPU computer. Additionally, we listed hash lengths of 10 schemes in the third column of Table 5. From Table 5, it's evident that schemes [19, 25, 28, 42] demonstrate lower computational complexity, and the generating time of hash sequence is 0.0064 s, 0.1018 s, 0.1077 s, and 0.2125 s, respectively. Conversely, schemes [38, 41] demonstrate high computational complexity, generating hash sequences in 0.9530 s and 1.3974 s. Additionally, hash sequence of schemes [19, 25] comprises 672 and 960 binary values, while schemes [23, 28, 30,

Table 5. Computational Complexity and Hash Sequence Length of Different Schemes

Scheme	Computational complexity	Hash length
[19]	0.2125 s	672 bits
[25]	0.1018 s	960 bits
[28]	0.1077 s	64 digits
[41]	0.9530 s	50 digits
[38]	1.3974 s	64 digits
[42]	0.0064 s	50 digits
[34]	0.4786 s	44 digits
[30]	0.4370 s	64 digits
[23]	0.3119 s	100 digits
[31]	0.3244 s	40 integers

34, 38, 41, 42] comprise 44 digits, 50 digits, 50 digits, 64 digits, 64 digits, 64 digits, and 100 digits, respectively. Meanwhile, the hash sequence of the scheme [31] comprises 40 integers.

6 Summary and Prospect

This survey presents a comprehensive review of the literature on perceptual hashing, describing the definition, steps, characters, and application scenarios. We delve into the latest advances in perceptual hashing and take a closer look at various types of perceptual hashing schemes, including perceptual image hashing, perceptual video hashing, and perceptual hashing for neural network model. Initially, researchers focused on proposing features using classic feature detectors. Subsequently, researchers began using machine learning or deep learning techniques to automatically extract features. However, in real application scenarios, multimedia data face countless editing and processing operations, and it is impossible to conduct supervised learning on all existing and possible future content-preserving manipulations. Perceptual hashing scheme based on unsupervised learning is intended to cope with unknown content-preserving manipulations.

Research on perceptual hashing is advancing steadily, with a variety of schemes and a wider range of authentication contents. However, there are still many limitations in existing schemes. For perceptual image hashing, when images contain substantial textual information, such as news, academic papers, or chat records, these schemes fail to effectively segment and process image and text components, leading to a significant decline in authentication performance. For perceptual video hashing, existing schemes take a long time to authenticate each video, which is difficult to meet practical application demands. Reducing the computational complexity and enhancing the efficiency of hash generation are urgent problems that need to be addressed. Additionally, videos often include audio information, which is overlooked by existing schemes. For neural network model hashing, with the increasing prevalence of large-scale neural network models, previous schemes struggle to perform effective authentication on these models. Compressing and representing these massive parameter models is an exceptionally challenging task. There are still many directions in this field that deserve further exploration.

(1) In practical application scenarios, the perceptual image hashing authentication system will face massive data, and the threshold used to judge whether perceptual identical or distinct will become particularly sensitive, which is prone to misjudgment. How to maintain excellent

performance in the face of massive data while improving the efficiency is a worthy research topic.

(2) Perceptual hashing schemes for multi-modal data is a topic worth studying. Current schemes are mainly aimed at one type of multimedia data, such as image, video, and so forth. However, data requiring authentication in real scenes are multi-modal, with both image and text, so the multi-modal scheme is worth further research.

(3) The multi-task or multi-function perceptual hashing scheme is also a worthy research topic. The current perceptual hashing scheme is to realize a single authentication function. Whether it can realize the recovery function after detecting the tampering of the multimedia data to be authenticated, or carry out multi-stage hierarchical content authentication, can be further studied and explored.

(4) Previously, perceptual image hashing generally focused on image content or images containing minimal text. However, in real-world scenarios, many images contain both visual and textual elements, such as news articles or chat records. Perceptual hashing for this kind of image presents a practical and worthwhile research direction.

(5) In addition to images, videos, and neural network models mentioned in this survey, the misuse of audio in the real-world is also increasingly prevalent, making perceptual audio hashing a field equally worth studying. Whether the perceptual content of original songs can be accurately authenticated after content-preserving manipulations such as clipping, frequency alteration, and speed playback. Furthermore, since many videos include audio information, this can be considered as a branch to complement perceptual video hashing, enabling more comprehensive authentication.

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